

Robustness of 3D U-Net Dose Prediction to Clinically Plausible CT Perturbations

OpenKBP Project

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Abstract

We evaluate the robustness of a 3D U-Net dose prediction model to five families of clinically plausible CT perturbations: acquisition noise, HU calibration shift, low-frequency bias field, resolution degradation, and dental metal artifacts. Unlike adversarial attacks that exploit worst-case gradient directions, these perturbations model realistic imaging variability encountered in radiotherapy workflows. Evaluated on 40 head-and-neck cancer patients at up to five severity levels per perturbation, the model demonstrates remarkable stability: four of five perturbation families cause less than 1.5% DVH degradation even at extreme amplitudes. Only resolution degradation produces substantial impact, with DVH degradation beginning at the mildest level (+1.5% at L0) and reaching +18.2% at L4. Bone-weighted calibration shift shows a late take-off at L5 (+11.2% DVH). These finer-grained severity sweeps identify specific threshold amplitudes for each perturbation, supporting targeted quality control recommendations for clinical deployment.

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1 Introduction

1.1 Clinical Motivation

CT imaging serves as the foundation of radiotherapy treatment planning. In clinical practice, CT images vary due to scanner hardware differences, acquisition protocols, reconstruction parameters, patient-specific artifacts (e.g., dental implants), and normal quantum noise. A dose prediction model deployed across clinical sites must produce stable predictions despite this inherent imaging variability.

1.2 Relationship to Adversarial Evaluation

A companion report (Report 1) evaluates the same model under adversarial attacks (FGSM, PGD) that exploit worst-case gradient directions to maximally degrade predictions. Those attacks revealed 21–30% dose score degradation at $\epsilon = 0.01$ (≈ 41 HU). This report takes a complementary approach: instead of worst-case perturbations, we apply *physically motivated* perturbations that model realistic CT imaging variability. This distinction is critical for clinical deployment—adversarial robustness measures theoretical vulnerability, while perturbation robustness measures practical reliability.

1.3 Model Overview

The evaluated model is a 3D U-Net with Squeeze-and-Excitation (SE) blocks, trained on the OpenKBP dataset for head-and-neck radiotherapy dose prediction:

- **Architecture:** 3D U-Net, 6 encoder / 5 decoder stages (64–512 filters), SE blocks on deep layers, InstanceNorm, residual connections
- **Input:** CT volume (128^3) + 10 structure masks
- **Output:** Predicted dose distribution (128^3)
- **Training:** Masked MAE loss, PTV weighting ($4.0\times$), data augmentation, mixed precision
- **Baseline performance:** DVH Score 2.535, Dose Score 3.731 Gy

2 Methods

2.1 Perturbation Families

We define five perturbation families spanning the major sources of CT imaging variability in radiotherapy. Each family is tested at multiple severity levels to identify degradation thresholds. P1–P3 and P5 use levels L1–L5 (mild to extreme); P4 (resolution) uses L0–L4 since degradation begins at the mildest settings. Table 1 summarizes all perturbations and their parameters.

Table 1: CT perturbation families with parameters at each severity level. All values in HU (Hounsfield Units) or voxels as indicated. L0 is only used for P4; L5 is not used for P4.

Perturbation	Physical Model	Levels (key parameters)
P1: Acq. Noise	Heteroscedastic Gaussian	L1: $\sigma_{\text{soft}/\text{bone}}=8/12$; L2: 15/25; L3: 30/50; L4: 60/100; L5: 100/160 HU
P2: Bone Shift	HU calibration error	L1: $\Delta_{\text{soft}/\text{bone}}=5/50$; L2: 10/100; L3: 25/250; L4: 50/500; L5: 100/1000 HU
P3: Bias Field	Low-freq. cosine harmonics	L1: 10; L2: 20; L3: 50; L4: 100; L5: 200 HU amplitude
P4: Resolution	Anisotropic Gaussian blur	L0: $\sigma_{z/xy}=0.5/0.25$; L1: 1.0/0.5; L2: 2.0/1.0; L3: 3.0/1.5; L4: 4.0/2.0 vox
P5: Dental Art.	Radial streak pattern	L1: 150 HU/8 streaks; L2: 300/12; L3: 500/16; L4: 800/20; L5: 1200/24

2.1.1 P1: Acquisition Noise

Models heteroscedastic quantum noise with tissue-dependent variance. The noise standard deviation is higher in bone (σ_{bone}) than in soft tissue (σ_{soft}), reflecting the physics of photon attenuation:

$$\text{CT}_{\text{perturbed}} = \text{CT} + \mathcal{N}(0, \sigma(x)^2), \quad \sigma(x) = \begin{cases} \sigma_{\text{bone}} & \text{if } \text{CT}(x) > 1500 \\ \sigma_{\text{soft}} & \text{otherwise} \end{cases} \quad (1)$$

where 1500 corresponds to the cortical bone threshold in OpenKBP coordinates (≈ 476 standard HU). At L1, noise magnitudes (8–12 HU) are within typical CT scanner noise ranges (10–50 HU); at L2, the 25 HU bone noise represents a noisier acquisition.

2.1.2 P2: Bone-Weighted HU Calibration Shift

Simulates systematic HU calibration drift that affects bone and soft tissue differently, as occurs with CT tube aging or cross-scanner variation. A sigmoid function provides smooth blending:

$$\Delta(x) = s \cdot \left[\Delta_{\text{soft}} + (\Delta_{\text{bone}} - \Delta_{\text{soft}}) \cdot \frac{1}{1 + e^{-(\text{CT}(x) - 1500)/200}} \right], \quad s \sim \text{Uniform}\{-1, +1\} \quad (2)$$

The random sign s models bidirectional calibration error. The sigmoid transition (center 1500, width 200) prevents discontinuities at tissue boundaries.

2.1.3 P3: Low-Frequency Bias Field

Models spatially smooth intensity variations from RF non-uniformity or scatter:

$$B(x, y, z) = A \cdot \frac{\sum_{k=1}^K w_k \prod_{d \in \{x, y, z\}} \cos(2\pi f_{k,d} \cdot d + \phi_{k,d})}{\max |B|} \quad (3)$$

where $K = 3$ harmonics with random frequencies $f \in [0.5, 2.5]$ cycles per FOV, phases $\phi \in [0, 2\pi]$, and weights $w \in [0.3, 1.0]$. The amplitude A controls the maximum HU deviation.

2.1.4 P4: Resolution Degradation

Models reduced spatial resolution from thicker CT slices or smoother reconstruction kernels via anisotropic Gaussian blurring:

$$\text{CT}_{\text{perturbed}} = \text{CT} * G(\sigma_z, \sigma_{xy}, \sigma_{xy}) \quad (4)$$

where G is a 3D Gaussian kernel applied with `scipy.ndimage.gaussian_filter`. The z-axis receives stronger blur than the xy-plane, reflecting the typical anisotropy of CT acquisitions where slice spacing exceeds in-plane resolution.

2.1.5 P5: Dental Metal Artifact

Simulates streak artifacts from dental implants anchored to the mandible centroid. The artifact consists of radial streaks in the axial plane with exponential radial decay and Gaussian z-spread:

$$\text{streak}(\theta) = \sum_{i=1}^N (-1)^i \exp\left(-\frac{(\theta - \theta_i)^2}{2w^2}\right), \quad w = \frac{\pi}{1.5N} \quad (5)$$

where N is the number of streaks and θ_i are random angles. The streak pattern is modulated by radial decay $e^{-r/40}$ and z-axis Gaussian decay ($\sigma_z = 3$ slices). A high-HU metal spot (3000–3500 HU) is placed at the mandible centroid within a 3-voxel radius.

2.2 Experimental Setup

- **Dataset:** 40 validation patients (pt_201–pt_240) from the OpenKBP dataset
- **Conditions:** 27 total (1 baseline + 5 perturbations $\times$ 4–5 levels each)
- **Seeding:** Deterministic (seed 42) for reproducibility; perturbation randomness is per-patient
- **CT normalization:** $[0, 4095] \rightarrow [0, 1]$ (12-bit range per OpenKBP specification)
- **Perturbation domain:** Applied in raw HU space before normalization
- **Hardware:** NVIDIA RTX 3090 GPU (RunPod), TensorFlow 2.18.0

2.3 Evaluation Metrics

- **Dose Score:** Voxel-wise MAE (Gy) within the `possible_dose_mask` region
- **DVH Score:** Mean absolute error of DVH statistics ($D_{0.1cc}$, mean dose for OARs; D_1 , D_{95} , D_{99} for PTVs) across all structures
- **Delta:** Absolute and percentage change from baseline for both metrics

Evaluation uses the OpenKBP `DoseEvaluator` class with unnormalized (Gy) dose values for both reference and prediction.

3 Results

3.1 Summary

Table 2 presents the complete results across all 27 conditions. The model achieves baseline performance of 3.731 Gy dose score and 2.535 DVH score.

Table 2: Dose and DVH scores across all perturbation conditions. Δ columns show change from baseline. Bold values indicate degradation $>5\%$.

Condition	Dose (Gy)	DVH	Δ Dose (%)	Δ DVH (%)
Baseline	3.731	2.535	—	—
P1_noise / L1	3.731	2.535	0.0	0.0
P1_noise / L2	3.733	2.538	0.1	0.1
P1_noise / L3	3.737	2.543	0.2	0.3
P1_noise / L4	3.742	2.559	0.3	1.0
P1_noise / L5	3.742	2.571	0.3	1.4
P2_bone_shift / L1	3.730	2.531	0.0	−0.1
P2_bone_shift / L2	3.740	2.544	0.3	0.4
P2_bone_shift / L3	3.751	2.519	0.5	−0.6
P2_bone_shift / L4	3.793	2.572	1.7	1.5
P2_bone_shift / L5	3.916	2.818	5.0	11.2
P3_bias_field / L1	3.732	2.535	0.0	0.0
P3_bias_field / L2	3.730	2.534	0.0	0.0
P3_bias_field / L3	3.736	2.534	0.2	0.0
P3_bias_field / L4	3.733	2.524	0.1	−0.4
P3_bias_field / L5	3.759	2.545	0.8	0.4
P4_resolution / L0	3.739	2.573	0.2	1.5
P4_resolution / L1	3.780	2.636	1.3	4.0
P4_resolution / L2	3.885	2.729	4.1	7.7
P4_resolution / L3	3.982	2.824	6.7	11.4
P4_resolution / L4	4.124	2.995	10.5	18.2
P5_dental / L1	3.729	2.541	0.0	0.3
P5_dental / L2	3.730	2.545	0.0	0.4
P5_dental / L3	3.730	2.552	0.0	0.7
P5_dental / L4	3.731	2.557	0.0	0.9
P5_dental / L5	3.732	2.555	0.0	0.8

3.2 Score Degradation Overview

Figure 1 shows the dose and DVH score degradation across all conditions. Resolution degradation (P4) is visually dominant; all other perturbations are near-zero.

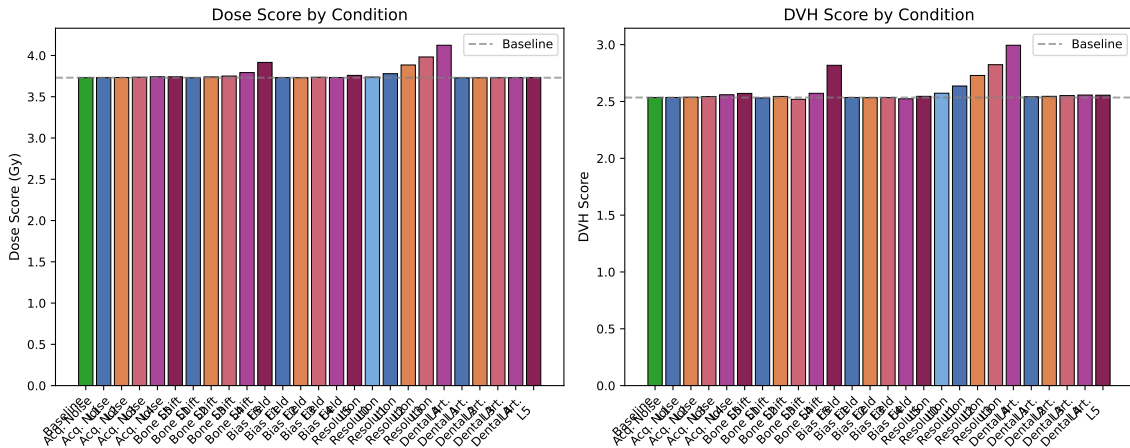


Figure 1: Dose score (left) and DVH score (right) across all perturbation conditions. P4 (resolution degradation) is the only perturbation producing visible degradation.

3.3 Degradation Curves

Figure 2 plots dose and DVH score as a function of perturbation severity for each perturbation family. The finer-grained severity sweep (up to 5 levels per perturbation) reveals distinct take-off behaviors:

- **P4 (Resolution):** Continuous, approximately linear degradation beginning at the mildest level (L0: +1.5% DVH) and reaching +18.2% DVH at L4. There is no threshold below which the model is invariant to spatial blur—even half-voxel Gaussian smoothing produces measurable impact.
- **P2 (Bone Shift):** Flat through L3 ($\Delta_{\text{bone}} \leq 250$ HU), then takes off at L4 (+1.5% DVH) with a sharp jump to L5 (+11.2% DVH, $\Delta_{\text{bone}} = 1000$ HU). This late, sudden onset indicates a threshold effect near $\Delta_{\text{bone}} \approx 500$ HU.
- **P1 (Noise), P3 (Bias Field), P5 (Dental):** All three remain essentially flat through their highest levels ($\leq 1.4\%$ DVH), confirming robust invariance to even extreme intensity-domain perturbations.

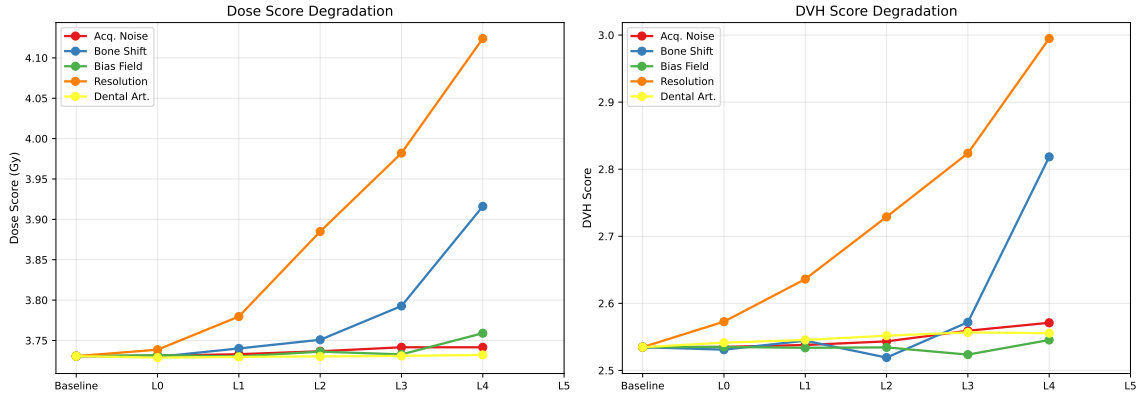


Figure 2: Degradation curves showing dose score (left) and DVH score (right) as perturbation severity increases from Baseline through up to 5 levels. P4 (resolution) degrades linearly from the mildest level; P2 (bone shift) shows a late take-off at L4–L5; P1, P3, and P5 remain flat.

3.4 Degradation Heatmap

Figure 3 provides a per-perturbation breakdown of dose and DVH score deltas, highlighting the contrast between P4 and the remaining perturbations.

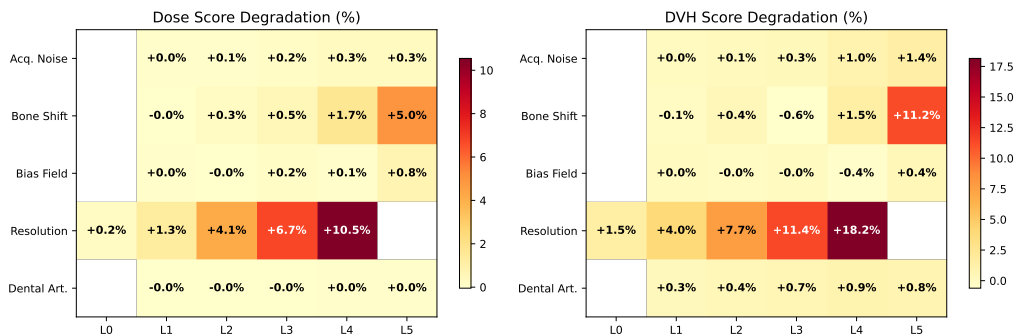


Figure 3: Heatmap of percentage degradation in dose score and DVH score by perturbation condition. P4 shows progressive degradation from L0 (+1.5% DVH) to L4 (+18.2% DVH); P2 shows a late take-off at L5 (+11.2% DVH). All other cells remain below 1.5%.

3.5 Per-Structure Analysis

Figure 4 shows the per-structure DVH metric errors under each perturbation. Resolution degradation particularly affects parotid glands, larynx, and mandible—structures with complex geometry where spatial blurring shifts dose-volume boundaries.

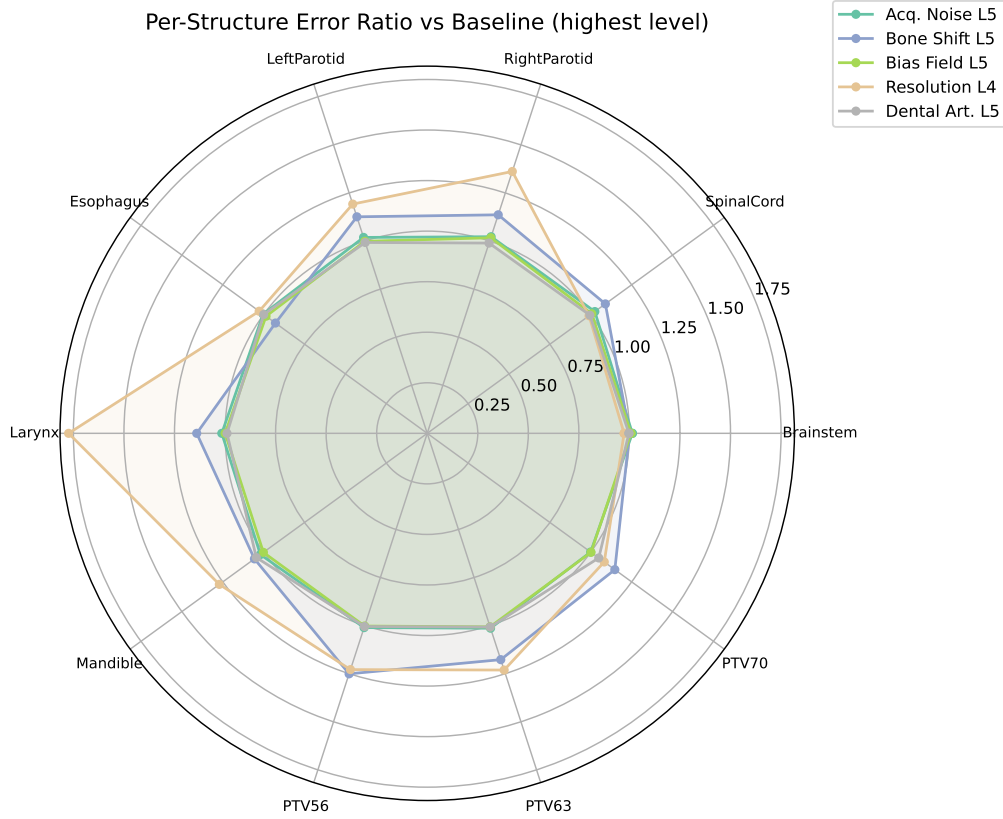


Figure 4: Radar chart of per-structure DVH errors across perturbation conditions. Baseline is shown as the reference ring; P4/L2 expands outward on structures with dose gradients near boundaries.

3.6 Visual Examples

Figures 5 and 6 show representative axial CT slices under each perturbation and the corresponding dose difference maps.

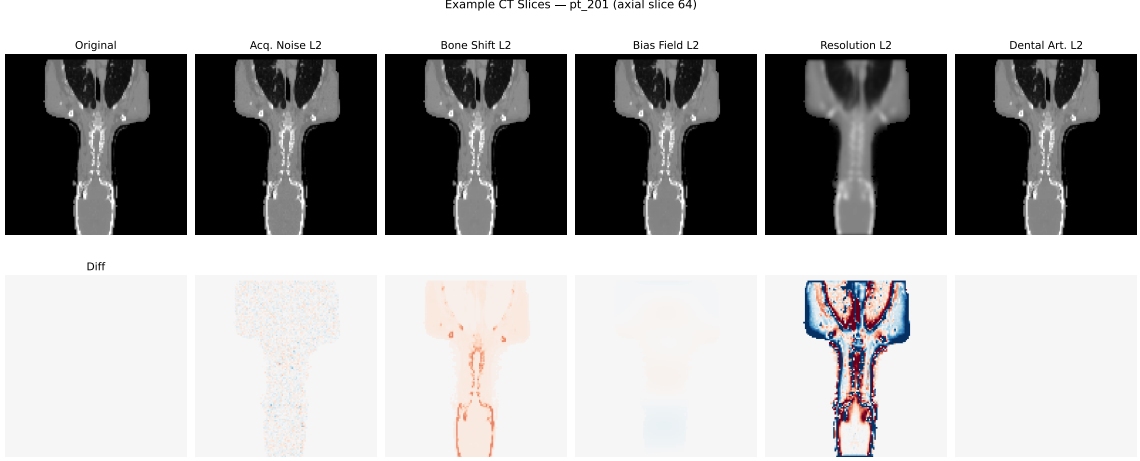


Figure 5: Axial CT slices (coronal view, head at top) for a representative patient under each perturbation at L2. Top row: perturbed CT; bottom row: difference from original. P1 (noise) and P3 (bias field) produce subtle changes; P2 (bone shift) shows intensity shifts in dense structures; P4 (resolution) visibly blurs tissue boundaries; P5 (dental) adds radial streaks near the mandible.

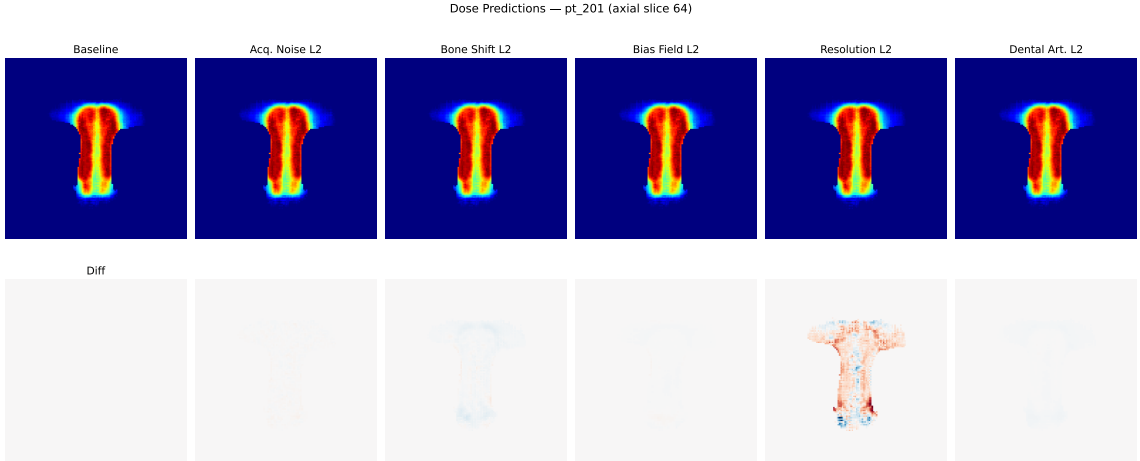


Figure 6: Dose predictions (top, coronal view) and difference maps (bottom, perturbed — baseline) for the same patient. P4 (resolution) produces spatially structured dose errors concentrated at organ boundaries, consistent with its dominant impact on DVH metrics.

4 Discussion

4.1 Key Findings

1. **Robust to most intensity-domain perturbations:** P1 (noise), P3 (bias field), and P5 (dental artifacts) cause $\leq 1.4\%$ DVH degradation even at their highest severity levels (L5), confirming that the model has learned dose-relevant features that are largely invariant to additive noise, smooth intensity fields, and localized metal artifacts.
2. **Resolution degradation: immediate, linear sensitivity:** P4 is the dominant perturbation, with DVH degradation beginning at the mildest level tested (L0: $\sigma_z = 0.5$ vox, $+1.5\%$ DVH) and climbing approximately linearly to $+18.2\%$ DVH at L4 ($\sigma_z = 4.0$ vox). There is no safe threshold below which blur is tolerable—even sub-voxel smoothing

degrades predictions. Blurring degrades the model’s ability to localize organ boundaries, which directly affects DVH metric computation.

3. **Bone shift: late threshold effect:** P2 remains flat through L3 ($\Delta_{\text{bone}} = 250$ HU) but takes off sharply at L4–L5, reaching +11.2% DVH at L5 ($\Delta_{\text{bone}} = 1000$ HU). This sudden onset suggests a threshold near $\Delta_{\text{bone}} \approx 500$ HU, well above typical calibration drift (10–50 HU), indicating the model is safe against realistic calibration errors.
4. **DVH more sensitive than dose score:** Resolution degradation causes +4.0% DVH degradation but only +1.3% dose score degradation at L1. DVH metrics, which depend on structure-specific dose statistics, are more sensitive to boundary shifts than voxel-wise MAE. This pattern holds across all levels.
5. **Structure-dependent vulnerability:** Under P4, parotid glands, larynx, and mandible show the largest DVH increases, likely because these structures have steep dose gradients at their boundaries. Structures in low-gradient regions (e.g., spinal cord) are less affected.

4.2 Comparison with Adversarial Results (Report 1)

The contrast between adversarial and physics-based perturbations is striking:

Table 3: Comparison of adversarial attacks (Report 1) vs. physics-based perturbations at similar HU magnitudes.

Perturbation Type	Approx. HU	Dose Δ	Nature
FGSM, $\epsilon = 0.01$	~ 41 HU	+30.4%	Worst-case gradient direction
PGD, $\epsilon = 0.01$	~ 41 HU	+21.3%	Iterative worst-case
P1 noise, L5	$\sigma = 100\text{--}160$ HU	+0.3%	Random Gaussian
P2 bone shift, L5	100–1000 HU	+5.0%	Systematic calibration
P3 bias field, L5	± 200 HU	+0.8%	Smooth spatial field

Even at extreme perturbation amplitudes (L5: $\sigma = 100\text{--}160$ HU for noise, $\Delta = 1000$ HU for bone shift), physically motivated perturbations cause at most +5.0% dose degradation (P2/L5), compared to +30% from FGSM at just 41 HU. At comparable HU magnitudes, the gap between adversarial and realistic impacts exceeds 100 $\times$. This confirms that the model’s adversarial vulnerability (Report 1) does not translate to practical fragility under real imaging conditions—real-world CT variability does not follow adversarial gradient directions.

4.3 Clinical Implications

- **CT spatial resolution matters most:** Clinical sites using thicker CT slices or smoother reconstruction kernels should expect degradation in dose prediction accuracy—even sub-voxel blur is detectable. Standardizing spatial resolution across sites would be the single most impactful quality control measure.
- **Intensity variations are well-tolerated:** Scanner noise, bias fields, and dental metal artifacts have negligible impact ($\leq 1.4\%$ DVH) even at extreme amplitudes, supporting cross-scanner deployment.
- **Calibration drift is safe within clinical ranges:** Bone-weighted HU shifts only become problematic above ~ 500 HU, far beyond typical inter-scanner calibration drift (10–50 HU). However, sites with severely miscalibrated scanners should be aware of the threshold.

- **No retraining needed for most variations:** The model’s inherent robustness to P1, P3, and P5 suggests that data augmentation or adversarial training is unnecessary for handling typical CT intensity variations.

4.4 Limitations

- **2D perturbation design:** P5 (dental artifact) applies 2D streak patterns extruded along z ; real metal artifacts have more complex 3D geometry.
- **Discrete severity levels:** While 4–5 levels per perturbation provide good threshold identification, continuous severity sweeps could reveal finer-grained threshold effects, particularly for P2 between L3 and L5.
- **Single model:** Results are specific to this architecture and training configuration; other models may have different robustness profiles.
- **DVH score limitations:** DVH metrics aggregate information across structures; focal dose errors in small regions may be masked by structure-level averaging.
- **Noise applied per-voxel:** P1 noise is i.i.d. per voxel; real CT noise has spatial correlations from reconstruction.

5 Conclusion

This evaluation demonstrates that the 3D U-Net dose prediction model is highly robust to clinically plausible CT imaging variations. Among five perturbation families spanning the major sources of CT variability:

1. **Three of five** perturbation families (noise, bias field, dental artifacts) cause $\leq 1.4\%$ DVH degradation even at extreme severity levels (L5), confirming robust invariance to intensity-domain variations.
2. **Resolution degradation** is the dominant sensitivity, with no safe threshold—degradation begins at L0 ($\sigma_z = 0.5$ vox, $+1.5\%$ DVH) and reaches $+18.2\%$ DVH at L4 ($\sigma_z = 4.0$ vox), identifying CT spatial resolution as the primary quality control factor.
3. **Bone-weighted calibration shift** shows a threshold effect, remaining flat through L3 ($\Delta_{\text{bone}} = 250$ HU) before taking off at L4–L5 (up to $+11.2\%$ DVH at $\Delta_{\text{bone}} = 1000$ HU). Typical clinical calibration drift is well below this threshold.
4. The $>100\times$ gap between adversarial and realistic perturbation impacts confirms that the model’s adversarial vulnerability (Report 1) does not translate to practical fragility under real imaging conditions.

These findings support the model’s suitability for multi-site clinical deployment, provided that CT spatial resolution is standardized or accounted for in the prediction pipeline.

6 Reproducibility

```
# Step 1: Generate perturbed CT volumes
cd open-kbp-modified/
python openkbp_hn_robustness/generate_perturbed_data.py

# Step 2: Run inference on all conditions (requires GPU + TF 2.18.0)
```

```
python openkbp_hn_robustness/run_inference.py
```

```
# Step 3: Evaluate metrics (CPU only)
```

```
python openkbp_hn_robustness/evaluate_metrics.py
```

```
# Step 4: Generate figures
```

```
python openkbp_hn_robustness/visualize_results.py
```

Configuration: `openkbp_hn_robustness/configs/default.yaml`

Acknowledgments

This work builds upon the OpenKBP Grand Challenge dataset and baseline implementation. Computation performed on RunPod cloud infrastructure with NVIDIA RTX 3090 GPU.